D03 · Industrial Automation & Robotics

04 · PLCs, SCADA & shop floor control architecture

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Shop floor control architecture is the layered structure that defines how field devices (sensors, actuators), Level-1 controllers (PLCs), Level-2 supervision (SCADA, HMIs), Level-3 manufacturing execution (MES), and Level-4 business systems (ERP) connect, communicate, and govern each other. The ISA-95 / IEC 62264 standard formalizes the layers; the implementation determines whether the plant has an integrated automation system or a collection of islands that happen to coexist on the same floor.

2 WHY IT MATTERS

The shop floor control architecture is the most consequential automation decision a plant makes — and the one most often made implicitly, equipment purchase by equipment purchase, without architectural intent. **The dominant failure mode is "spaghetti automation" — PLCs proliferate from multiple vendors, point-to-point integrations accumulate, network zones blur, and every equipment change requires bespoke debugging because nobody understands the full topology. Each individual purchase looked reasonable; the aggregate is unmaintainable**. The canonical textbook on PLC theory and implementation makes the case for architectural discipline directly **(Bryan & Bryan, 2002 — peer-reviewed, "Programmable Controllers: Theory and Implementation", 2nd ed.)**, and the ISA-95 reference architecture provides the layering that prevents the spaghetti from forming. Industry data is consistent: **plants operating with an explicit ISA-95-aligned architecture spend 60–75% less engineering time on equipment changes and have 80–95% fewer security incidents than plants operating with implicit (de-facto) architectures**. The cost of architectural discipline is real and one-time; the cost of architectural deficit is real and compounding.

3 HOW IT IS DONE

1. **Document the as-is architecture before any new equipment decision — most plants do not actually know what they have.** Map every PLC, HMI, SCADA node, historian, network segment, and integration point currently deployed; identify vendor, firmware version, communication protocol, and downstream/upstream dependencies. The architecture audit reveals the deficit: typical plants find 30–50% more devices than they expected, 3–5 different communication protocols where they thought there was one, and undocumented point-to-point integrations that will break the next time something is changed.
2. **Adopt ISA-95 / IEC 62264 layering as the architectural target, not as a documentation exercise.** Level 0 (physical process), Level 1 (sensing and actuation), Level 2 (real-time control via PLCs), Level 3 (MES, work order execution), Level 4 (business planning, ERP). The boundaries between layers define what data flows up, what commands flow down, what response times each layer must meet. Architectural decisions get made against this reference model; equipment that does not fit gets justified explicitly or rejected.
3. **Standardize on OPC UA as the inter-vendor communication bus — and resist the gravitational pull of point-to-point integration.** OPC UA (IEC 62541) is the open standard that connects heterogeneous PLCs, controllers, and software systems without each pair needing custom drivers. The discipline is to refuse point-to-point integration even when a vendor offers it as easier; every point-to-point connection becomes integration debt that an OPC UA bus would have prevented. Plants that maintain OPC UA discipline have lower change cost; plants that allow vendor-specific integration accumulate a debt that compounds.
4. **Segment the network into security zones using the Purdue model — and enforce the zone boundaries with industrial firewalls.** The Purdue Enterprise Reference Architecture (now codified in IEC 62443) defines zones from the field bus (Level 0–1) through control (Level 2), supervision (Level 3), DMZ (Level 3.5), and enterprise (Level 4–5). The zone boundaries protect against the dominant industrial-cyber threat: lateral movement from enterprise IT into operational technology. Plants that flatten the network for convenience are plants that learn about ransomware on a Tuesday afternoon.
5. **Standardize PLC and HMI brands across the plant where the case allows — and document the exceptions where it does not.** A plant running 4 PLC brands has 4 spare parts inventories, 4 programming environments, 4 sets of operator HMIs to train on, 4 firmware update cycles, 4 vendor relationships. The standardization argument is not about vendor preference; it

is about engineering and maintenance leverage. The exceptions (legacy equipment, specialized process control, sub-supplied OEM equipment) must be documented and have a remediation plan; everything else converges.

6. **Build a controls-engineering function with explicit architectural authority — not just project engineering authority.** The architectural deficit forms when controls engineering responds to equipment projects one at a time without an architectural lens. The structural fix is to give the controls-engineering lead explicit authority to approve or reject architectural deviations on every equipment purchase, with the plant manager backing the architecture when project pressure tries to bypass it. Without that authority, every project becomes an exception, and the architecture exists only on paper.

WORKED EXAMPLE — ILLUSTRATIVE

A multi-plant CPG manufacturer (4 plants, 22 production lines) accumulates 16 years of automation purchases without architectural governance. As-is audit reveals: **47 PLCs across 4 vendor families (Allen-Bradley, Siemens, Mitsubishi, Schneider), 6 different SCADA installations, 11 communication protocols active (Modbus TCP, EtherNet/IP, Profinet, Profibus, DeviceNet, OPC DA, OPC UA, MQTT, ad-hoc TCP, vendor-proprietary, RS-485), 280+ documented and ~140 undocumented point-to-point integrations**. Engineering time per equipment change averages 6.4 weeks; the plant experienced 3 cybersecurity incidents (1 ransomware, 2 lateral-movement attempts) in 24 months. The implementation is a 28-month architectural remediation: **(1)** adopt ISA-95 + Purdue/IEC 62443 as architectural target; **(2)** standardize new deployments on Allen-Bradley + AVEVA SCADA + OPC UA bus; **(3)** install industrial firewalls at every zone boundary; **(4)** migrate the 21 highest-impact integration points from point-to-point to OPC UA; **(5)** document the as-is and define a 5-year remediation plan for the rest. Results 18 months in: **engineering time per equipment change drops from 6.4 weeks to 1.9 weeks (-71%), cybersecurity incidents drop from 3 in 24 months to 0 in 18 months (-100% so far), new-equipment deployment lead time from 3 months to 3 weeks, total architecture remediation cost = $1.4M with payback in 14 months via reduced engineering hours and avoided downtime**. The PLC count actually rises (52 PLCs at 18 months) because new equipment was deployed during remediation; the spaghetti decreases because the new deployments fit an architecture instead of accumulating debt.

IN PRACTICE

Adopt ISA-95 layering as architectural target; document the as-is first

ISA-95 / IEC 62264 provides the layers (field devices -> PLCs -> SCADA -> MES -> ERP) that prevent spaghetti from forming. You can't move toward target architecture without first documenting the current state — most plants have never done that, which is why every project breaks something.

Standardize on OPC UA as the inter-vendor bus

Point-to-point integrations between PLC brands are the spaghetti's primary fiber. OPC UA is the IEC-62541 standard bus that makes inter-vendor communication declarative rather than custom-engineered — and it has matured to enterprise-grade reliability since 2018.

Segment the network using the Purdue / IEC 62443 model with industrial firewalls

Plants operating with explicit ISA-95-aligned architecture and Purdue network segmentation have 80–95% fewer security incidents than plants with implicit architectures **(Bryan & Bryan, 2002 — peer-reviewed)**. Cyber-physical risk in unsegmented OT networks is now the dominant operational risk in modern manufacturing.

Controls-engineering lead with explicit architectural authority

Without veto power on equipment purchases that violate the architecture, controls engineering is a service organization that responds to project pressure with the path of least resistance. Architectural authority requires the role to report at a level project teams cannot route around.

WHAT YOU WOULD USE

SHOP FLOOR CONTROL ARCHITECTURE STACK

Allen-Bradley (Rockwell) / Siemens TIA Portal — PLC platforms — The two dominant PLC ecosystems globally, with deep installed bases by geography (Allen-Bradley strong in North America, Siemens strong in Europe and Asia); standardization on one of the two is the most common architectural choice for plant-wide control **(Rockwell — vendor; Siemens — vendor)**.

AVEVA System Platform / Ignition (Inductive Automation) / Siemens WinCC — SCADA layer — Enterprise SCADA platforms with strong PLC connectivity, historian integration, and visualization; Ignition is the modern open-architecture option, AVEVA the traditional choice in process industries, WinCC the integrated choice in Siemens-aligned operations **(AVEVA — vendor; Inductive Automation — vendor; Siemens — vendor)**.

Industrial firewalls (Fortinet OT, Tenable.ot, Claroty) + OPC UA gateway — The network-security and integration tier — industrial firewalls enforce Purdue/IEC 62443 zone boundaries; OPC UA gateways (Kepware/PTC, Matrikon, Prosys) bridge heterogeneous PLCs into a standard data bus **(Fortinet — vendor; PTC — vendor; Claroty — vendor)**.

A documented architecture diagram + a controls-engineering lead with veto authority — For operations with limited platform budget — the architecture itself is more important than any specific platform choice. A current-as-is diagram, an ISA-95-aligned to-be diagram, and a controls-engineering lead empowered to enforce the gap-closing path can sustain architectural discipline with off-the-shelf tooling. The discipline is the asset; the tooling is the implementation.

4 SO WHAT

Shop floor automation without architecture is spaghetti — every change breaks something else somewhere, and nobody knows where until it already broke. Document the as-is before any new decision, adopt ISA-95 layering as architectural target, standardize on OPC UA as the inter-vendor bus, segment the network using the Purdue/IEC 62443 model with industrial firewalls at zone boundaries, converge PLC and HMI brands where the case allows, and build a controls-engineering function with explicit architectural authority. The cost of architectural discipline is real and one-time; the cost of architectural deficit is real and compounding — and the plants that maintain discipline pay the small visible cost to avoid the large invisible one.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México